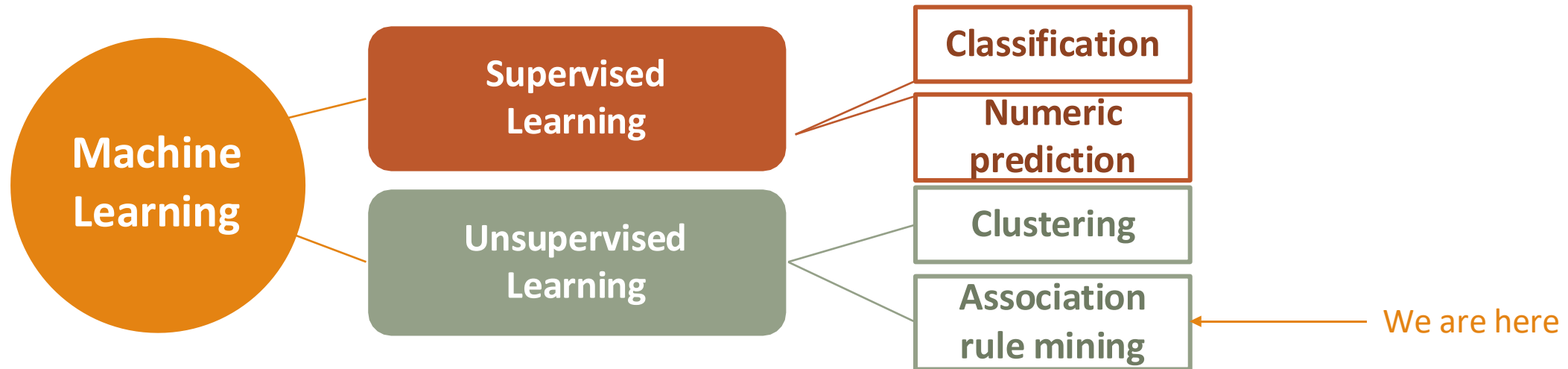


Lecture 10:

Unsupervised Learning:

Association Rule Mining

Data Mining Tasks: Recap



Outline

- Finding Patterns – Market Basket Analysis
- Understanding Association Rules
- Apriori Algorithm for Association Rule Learning
- Measuring Rule Interest – Support, Confidence, and Lift

Market Basket Analysis

- **Unsupervised learning-Association rule mining**
 - A classic case: Diaper and Beer



Market Basket Analysis

- **Unsupervised learning-Association rule mining**

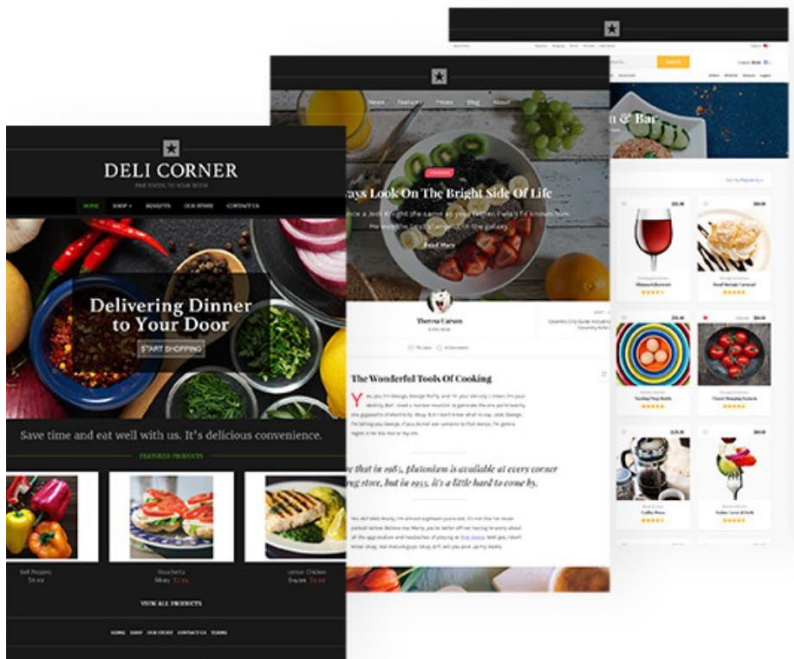
The screenshot shows the Amazon.com interface for a book. At the top, the Amazon logo and navigation links are visible. The search bar contains the text 'Books'. Below the search bar, the book 'Data Mining: Practical Machine Learning Tools and Techniques [Paperback]' by Ian H. Witten and Elke Frank is displayed. The book cover is green with the title 'DATA MINING' in large yellow letters. The authors' names are listed below the title. The price is shown as \$43.44, with a list price of \$68.95 crossed out. A 'Click to LOOK INSIDE!' button is present. The book is marked as 'In Stock' and 'Ships from and sold by Amazon.com'. A promotional banner at the bottom encourages joining Amazon Student for free two-day shipping.

The screenshot shows the 'Frequently Bought Together' section. It displays three book covers: 'Data Mining: Practical Machine Learning Tools and Techniques, Second Edition' (green cover), 'Data Mining: Concepts and Techniques, Second Edition' (blue cover), and 'Handbook of Statistical Analysis and Data Mining Applications' (black cover). The price for all three is listed as \$178.50. There are buttons for 'Add all three to Cart' and 'Add all three to Wish List'. A link to 'Show availability and shipping details' is also present. Below the books, a list of items is shown with checkboxes and descriptions: 'This item: Data Mining: Practical Machine Learning Tools and Techniques, Second Edition (Morgan Kaufmann Series in Data Mining)', 'Data Mining: Concepts and Techniques, Second Edition (The Morgan Kaufmann Series in Data Mining)', and 'Handbook of Statistical Analysis and Data Mining Applications by Robert Nisbet Hardcover \$80.59'.

Promotions to
retain customers
& increase sales

Market Basket Analysis

- **Web Log analysis:** Web sequential patterns are very useful to better structure a company's website for providing easier access to the most popular links.



- We can use association rules to discover the pages which are visited together even if they are not directly connected.
- The rules can be used for restructuring Web sites by adding links between those pages which are visited together.

Understanding Association Rules

- Association rule mining is **unsupervised learning**
 - No need for the algorithm to be trained
 - Data does not need to be labeled ahead of time
 - The program is simply unleashed on a dataset in the hope that interesting associations are found
 - No evaluation

Understanding Association Rules

- The following table shows five transactions in a hospital's gift shop:

Transaction number	Purchased items
1	<i>{flowers, get well card, soda}</i>
2	<i>{plush toy bear, flowers, balloons, candy bar}</i>
3	<i>{get well card, candy bar, flowers}</i>
4	<i>{plush toy bear, balloons, soda}</i>
5	<i>{flowers, get well card, soda}</i>

Understanding Association Rules

- The building blocks of an association rule mining are the **items** that may appear in any given transaction.
- **Transactions are specified in terms of itemset.**



Understanding Association Rules

- The result of a market basket analysis is a collection of **association rules** specify patterns found in the relationships among items in the itemsets.
- Association rules are always composed from subsets of itemsets and are denoted by relating one itemset on the ***left-hand side (LHS)*** of the rule to another itemset on the ***right-hand side (RHS)*** of the rule.
- The ***LHS is the condition*** that needs to be met in order to trigger the rule, and the ***RHS is the expected result*** of meeting that condition.



Understanding Association Rules

$\{\text{flowers, get well card}\} \longrightarrow \{\text{soda}\}$

- If *flowers* and *get well card* are purchased together, then *soda* is also likely to be purchased. In other words, "*flowers* and *get well card* imply *soda*."

Understanding Association Rules

- Association rule analysis: search for interesting connections among a very large number of elements.
 - Human beings are capable of such insight quite intuitively
 - It often takes expert-level knowledge or a great deal of experience to do what a rule learning algorithm can do in minutes or even seconds.
 - Additionally, some datasets are simply too large and complex

The Apriori Algorithm for Association Rule Learning

- The most-widely used approach for efficiently searching large databases for rules is known as **Apriori**.
- The following table shows five transactions in a hospital's gift shop:

Transaction number	Purchased items
1	<i>{flowers, get well card, soda}</i>
2	<i>{plush toy bear, flowers, balloons, candy bar}</i>
3	<i>{get well card, candy bar, flowers}</i>
4	<i>{plush toy bear, balloons, soda}</i>
5	<i>{flowers, get well card, soda}</i>

The Apriori Algorithm for Association Rule Learning

- The Apriori algorithm uses statistical measures of an itemset's "**interestingness**" to locate association rules in much larger transaction databases.
- Whether or not an association rule is deemed interesting is determined by two statistical measures: **support** and **confidence** measures.

The Apriori Algorithm for Association Rule Learning

- The **support** of an itemset or rule measures how frequently it occurs in the data.
 - Support for the itemset X : $\text{support}(X) = \frac{\text{count}(X)}{N}$
 - N is the number of transactions in the database and $\text{count}(X)$ is the number of transactions containing itemset X
 - Support for a single item:
 - The support for $\{candy\ bar\}$ is $2 / 5 = 0.4$
 - Support for itemset:
 - $\{get\ well\ card, flowers\}$ has support of $3 / 5 = 0.6$ in the hospital gift shop data
 - The support for $\{get\ well\ card\} \rightarrow \{flowers\}$ is also 0.6
 - The support for $\{flowers\} \rightarrow \{get\ well\ card\}$ is also 0.6

The Apriori Algorithm for Association Rule Learning

- A rule's **confidence** is defined as the support of the itemset containing both X and Y divided by the support of the itemset containing only X :

$$\text{confidence}(X \rightarrow Y) = \frac{\text{support}(X, Y)}{\text{support}(X)}$$

- The confidence tells us the proportion of transactions where the presence of item or itemset X results in the presence of item or itemset Y

The Apriori Algorithm for Association Rule Learning

- **Confidence that X leads to Y is not the same as the confidence that Y leads to X**
 - The confidence of $\{flowers\} \rightarrow \{get\ well\ card\}$ is $0.6 / 0.8 = 0.75$
 - The confidence of $\{get\ well\ card\} \rightarrow \{flowers\}$ is $0.6 / 0.6 = 1.0$
 - a purchase involving flowers is accompanied by a purchase of a get well card 75 percent of the time, while a purchase of a get well card is associated with flowers 100 percent of the time.

Transaction number	Purchased items
1	$\{flowers, get\ well\ card, soda\}$
2	$\{plush\ toy\ bear, flowers, balloons, candy\ bar\}$
3	$\{get\ well\ card, candy\ bar, flowers\}$
4	$\{plush\ toy\ bear, balloons, soda\}$
5	$\{flowers, get\ well\ card, soda\}$

High confidence value means if a customer bought the items on the left hand side, that customer also has a high chance to buy the items on the right hand side.

The Apriori Algorithm for Association Rule Learning

- Rules like $\{get\ well\ card\} \rightarrow \{flowers\}$ are known as **strong rules (interesting rules)**, because they have both high support and confidence.
 - Support value: 0.6
 - Confidence value: 1.0
- The way to find more strong rules would be to examine ***every possible combination of the items*** in the gift shop, measure the support and confidence value, and report back only those rules that meet certain levels of interest.

The Apriori Algorithm for Association Rule Learning

- Apriori algorithm creates rules in two phase:
 1. Identifying all the itemsets that meet a minimum support threshold.
 2. Creating rules from these frequent itemsets using those meeting a minimum confidence threshold.

Apriori Algorithm

- Consider a supermarket scenario

- Sell five items: Onion, Burger, Potato, Milk, Beer.

- The database consists of six transactions where 1 represents the presence of the item and 0 the absence.

Transaction ID	Onion	Potato	Burger	Milk	Beer
t_1	1	1	1	0	0
t_2	0	1	1	1	0
t_3	0	0	0	1	1
t_4	1	1	0	1	0
t_5	1	1	1	0	1
t_6	1	1	1	1	0

The Apriori Algorithm makes the following assumptions.

- All subsets of a frequent itemset should be frequent.
- The itemset containing infrequent items should be infrequent.
- Set a threshold support level. In our case, we shall fix it at 50%.

Apriori Algorithm

- **Step 1.** Create a frequency table of all the items that occur in all the transactions.
- **Step 2.** Here, support threshold is 50%, hence only those items are **frequent** which occur in equal, or more than three transactions and such items are Onion(O), Potato(P), Burger(B), and Milk(M).
- **Step 3.** make all the possible pairs of the significant items (the order doesn't matter)

1-item itemsets

Item	Frequency (No. of transactions)
{ Onion(O) }	4
{ Potato(P) }	5
{ Burger(B) }	4
{ Milk(M) }	4
{ Beer(Be) }	2

Generate frequent items



1-item itemsets that meet the minimum support threshold

Item	Frequency (No. of transactions)
{ Onion(O) }	4
{ Potato(P) }	5
{ Burger(B) }	4
{ Milk(M) }	4



2-item itemsets

Itemset	Frequency (No. of transactions)
{ OP }	4
{ OB }	3
{ OM }	2
{ PB }	4
{ PM }	3
{ BM }	2

- **Step 4.** Only those itemsets are significant which cross the support threshold (OP, OB, PB, and PM)
- **Step 5.** Now we would like to look for 3-item itemsets that are purchased together. We will use the item sets found in step 4 and create 3-item itemsets.
- **Step 6.** Applying the threshold rule again, we find that OPB is the only frequent 3-item itemset.

2-item itemsets

Itemset	Frequency (No. of transactions)
{ OP }	4
{ OB }	3
{ OM }	2
{ PB }	4
{ PM }	3
{ BM }	2

2-item itemsets that meet the minimum support threshold

Itemset	Frequency (No. of transactions)
{ OP }	4
{ OB }	3
{ OM }	2
{ PB }	4
{ PM }	3
{ BM }	2

3-item itemsets

Itemset	Frequency (No. of transactions)
{ OPB }	4

The Apriori Algorithm for Association Rule Learning

- Frequent itemsets
 - $\{O\}$, $\{P\}$, $\{B\}$, $\{M\}$, $\{OP\}$, $\{OB\}$, $\{PB\}$, $\{PM\}$, and $\{OPB\}$
- From frequent itemsets to association rules
 - $\{O\} \rightarrow \{P\}$, $\{P\} \rightarrow \{O\}$
 - $\{O\} \rightarrow \{B\}$, $\{B\} \rightarrow \{O\}$
 - $\{P\} \rightarrow \{B\}$, $\{B\} \rightarrow \{P\}$
 - $\{P\} \rightarrow \{M\}$, $\{M\} \rightarrow \{P\}$
 - $\{O,P\} \rightarrow \{B\}$, $\{B,P\} \rightarrow \{O\}$, $\{O,B\} \rightarrow \{P\}$, $\{B\} \rightarrow \{O,P\}$, $\{O\} \rightarrow \{B,P\}$, $\{P\} \rightarrow \{O,B\}$
- Exam the confidence value to generate **strong rules**

The Apriori Algorithm for Association Rule Learning

- The first phase occurs in multiple iterations.
- Each successive iteration involves evaluating the support of a set of increasingly large itemsets.
 - Iteration 1 involves evaluating the set of 1-item itemsets
 - Iteration 2 evaluates 2-item itemsets, and so on.
- The result of each iteration i is a set of all the frequent i -item itemsets that meet the minimum support threshold.
- All the frequent itemsets from iteration i are combined in order to generate candidate itemsets for the evaluation in iteration $i + 1$.

The Apriori Algorithm for Association Rule Learning

- The second phase of the Apriori algorithm may begin.
- Given the set of frequent itemsets, association rules are generated from all possible subsets.
 - For instance, $\{A, B\}$ would result in candidate rules for $\{A\} \rightarrow \{B\}$ and $\{B\} \rightarrow \{A\}$.
- These are evaluated against a minimum confidence threshold, and any rule that does not meet the desired confidence level is eliminated.

Identifying Frequently Purchased Groceries with Association Rules

- Market basket analysis on grocery data
 - We will utilize the purchase data collected from one month of operation at a real-world grocery store. The data contains 9,835 transactions.
 - Transactional data is stored in a slightly different format than that we used previously. Transactional data is a more free form.
 - Each row in the data specifies a single transaction.
 - Rather than having a set number of features, each record comprises a comma-separated list of any number of items, from one to many.
 - The items may differ from example to example

Identifying Frequently Purchased Groceries with Association Rules

- The first five rows of the raw grocery.csv file are as follows:

	A	B	C	D	E
1	citrus fruit	semi-finished bread	margarine	ready soups	
2	tropical fruit	yogurt	coffee		
3	whole milk				
4	pip fruit	yogurt	cream cheese	meat spreads	
5	other vegetables	whole milk	condensed milk	long life bakery product	

- These lines indicate five separate grocery store transactions.

Identifying Frequently Purchased Groceries with Association Rules

```
# Show the number of transactions  
len(groceries)
```

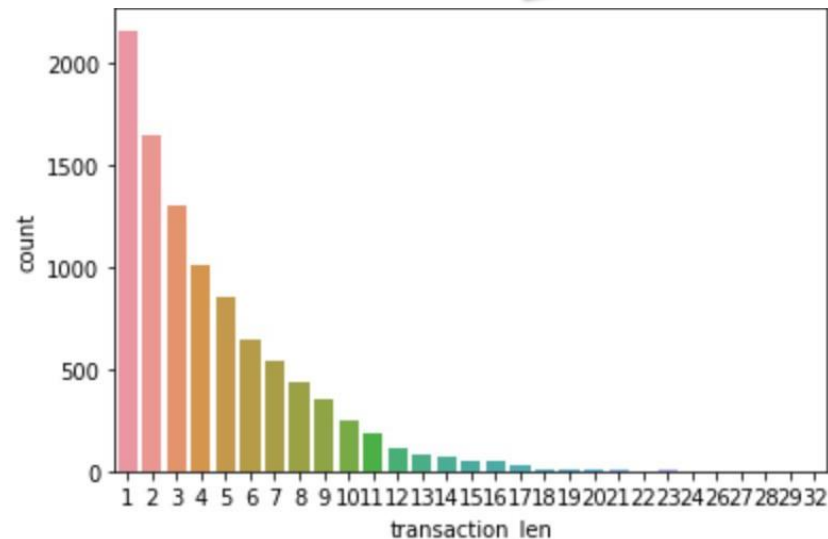
9835

The output 9835 rows refers to the number of transactions

```
# Show the number of unique items  
flat_list = np.concatenate(groceries).flat  
unique_item, item_count = np.unique(flat_list, return_counts=True)  
print("number of unique items:", len(unique_item))
```

number of unique items: 169

The output 169 columns refers to the 169 different items that might appear in someone's grocery basket.



A total of 2,159 transactions contained only a single item, while one transaction had 32 items

items	
whole milk	2513
other vegetables	1903
rolls/buns	1809
soda	1715
yogurt	1372
...	
bags	4
kitchen utensil	4
preservation products	2
baby food	1
sound storage medium	1
Length: 169, dtype: int64	

Identifying Frequently Purchased Groceries with Association Rules

- `groceries_rules = apriori(groceries, min_support = 0.01, min_confidence = 0.3, max_length=2)`
 - Find the support and confidence parameters that produce a reasonable number of association rules
 - Set these levels too high: find no rules or rules that are too generic to be very useful
 - Set the threshold too low: an unwieldy number of rules; the operation might take a very long time or run out of memory during the learning phase.

Identifying Frequently Purchased Groceries with Association Rules

- One way to approach the problem of setting a minimum support threshold
 - Think about the smallest number of transactions you would need before you would consider a pattern interesting.
 - For instance, you could argue that if an item is purchased twice a day (about 60 times in a month of data), it may be an interesting pattern.
 - 60 out of 9,835 equals 0.006
- Setting the minimum confidence involves a delicate balance.
 - Confidence is too low: overwhelmed with a large number of unreliable rules
 - set confidence too high: limited to the rules that are obvious or inevitable

Identifying Frequently Purchased Groceries with Association Rules

- The first rule can be read in plain language as, "if a customer buys butter, he/she will also buy whole milk."

```
# Display the top 5 rules sorted by confidence levels
all_rules.sort_values(by='confidence', ascending=False)[:5]
```

	LHS	RHS	support	confidence	lift
9	butter	whole milk	0.027555	0.497248	1.946053
22	curd	whole milk	0.026131	0.490458	1.919481
27	domestic eggs	whole milk	0.029995	0.472756	1.850203
44	onions	other vegetables	0.014235	0.459016	2.372268
66	whipped/sour cream	whole milk	0.032232	0.449645	1.759754

Measures

- Support for association rule $X \rightarrow Y$: $support(X, Y) = \frac{count(X, Y)}{N}$
- Confidence is defined as the support of the itemset containing both X and Y divided by the support of the itemset containing only X :
$$confidence(X \rightarrow Y) = \frac{support(X, Y)}{support(X)}$$
- The lift of a rule measures: how much more likely one item or itemset is purchased relative to its typical rate of purchase, given that you know another item or itemset has been purchased.
 - Unlike confidence where the item order matters, $lift(X \rightarrow Y)$ is the same as $lift(Y \rightarrow X)$.

$$lift(X \rightarrow Y) = \frac{confidence(X \rightarrow Y)}{support(Y)} = \frac{support(X, Y)}{support(X) * support(Y)}$$

Measures

$$\text{Confidence}(X \rightarrow Y) = P(Y \mid X)$$

$$\text{Lift}(X \rightarrow Y) = \frac{P(Y \mid X)}{P(Y)} = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

- **Lift = 1** → X and Y are **independent**
- **Lift > 1** → **Positive association** (X increases likelihood of Y)
- **Lift < 1** → **Negative association** (X decreases likelihood of Y)

Feasibility vs. Optimality

•Support & Confidence (Apriori):

- Ensure **feasible rules**
- Filter out rare or unreliable patterns

•Lift:

- Identifies **interesting/optimal rules**
- Highlights **true associations beyond chance**

Measures

Applications of Lift (Beyond Association Rules)

- ❑ **Marketing:** campaign effectiveness vs baseline response
- ❑ **A/B Testing:** measure improvement over control
- ❑ **Risk Modeling:** compare observed vs expected outcomes
- ❑ **Recommender Systems:** detect meaningful co-preferences
- ❑ **Healthcare:** identify significant co-occurring conditions

Leverage (Absolute Difference) vs. Lift

$$\text{Leverage}(X \rightarrow Y) = P(X) [P(Y | X) - P(Y)]$$

- Difference between actual probability of Y given X and baseline probability of Y, scaled by P(X)
- Captures absolute gain (or loss) due to X

Leverage Interpretation:

= 0 → no change (independence)

> 0 → X increases probability of Y

< 0 → X decreases the probability of Y

Identifying Frequently Purchased Groceries with Association Rules

LHS	RHS	support	confidence	lift
butter	whole milk	0.027555	0.497248	1.946053

0.055414 (545/9835) 0.255516 (2513/9835)

- Support: 0.027555 (271/9835)
 - The probability of buying butter and whole milk together is 0.027555
 - 2.755% transactions include both butter and whole milk. This rule covers 2.755% transactions.
- Confidence: 0.497248 (271/545)
 - When customers bought butter, they will also buy whole milk about 49.7248% of the time.
 - Among transactions including butter, 49.7248% of them also include whole milk.
- Lift: 1.946053 ($0.027555 / (0.055414 * 0.255516)$)
 - Butter and whole milk have **positive effect (because lift value greater than 1)** on each other.
 - Purchasing butter increases the probability of buying whole milk by 1.946 times.
 - Purchasing whole milk increases the probability of buying butter by 1.946 times.

Identifying Frequently Purchased Groceries with Association Rules

In spite of the fact that the confidence and lift are high, does $\{\text{butter}\} \rightarrow \{\text{whole milk}\}$ seem like a very useful rule?

Identifying Frequently Purchased Groceries with Association Rules

- A common approach is to take the association rules and divide them into the following three categories:
 - Actionable
 - Trivial
 - Inexplicable
- Obviously, the goal of a market basket analysis is to find actionable rules that provide a clear and useful insight.
 - Some rules are clear, others are useful
 - it is less common to find a combination of both of these factors.

Identifying Frequently Purchased Groceries with Association Rules

- **Trivial rules** include any rules that are so obvious that they are not worth mentioning
 - They are clear, but not useful.
 - {diapers} $\rightarrow$ {formula}
- Rules are **inexplicable** if the connection between the items is so unclear that figuring out how to use the information is impossible or nearly impossible.
 - Simply be a random pattern in the data

Identifying Frequently Purchased Groceries with Association Rules

- The best rules are hidden gems—those undiscovered insights into patterns that seem obvious once discovered.

Identifying Frequently Purchased Groceries with Association Rules

```
# Display all rules sorted by confidence levels
all_rules.sort_values(by = 'lift', ascending = False)
```

40	margarine	whole milk	0.024199	0.413194	1.617098
24	dessert	other vegetables	0.011591	0.312329	1.614164
54	yogurt	other vegetables	0.043416	0.311224	1.608457
36	hard cheese	whole milk	0.010066	0.410788	1.607682
13	chicken	whole milk	0.017590	0.409953	1.604411
3	berries	other vegetables	0.010269	0.308869	1.596280
67	white bread	whole milk	0.017082	0.405797	1.588147
2	beef	whole milk	0.021251	0.405039	1.585180
64	tropical fruit	whole milk	0.042298	0.403101	1.577595
43	oil	whole milk	0.011286	0.402174	1.573968

Identifying Frequently Purchased Groceries with Association Rules

- Suppose the marketing team is excited about the possibilities of creating an advertisement to promote vegetables.
- Before finalizing the campaign, however, they ask you to investigate whether vegetables are often purchased with other items.
- To answer this question, we'll need to find all the rules that include vegetables in some form.

Identifying Frequently Purchased Groceries with Association Rules

```
# Find and display rules containing "other vegetables" on the left-hand side
all_rules[all_rules['LHS'].str.contains("other vegetables")]
```

	LHS	RHS	support	confidence	lift
12	other vegetables, root vegetables	whole milk	0.023183	0.489270	1.914833
14	other vegetables, yogurt	whole milk	0.022267	0.512881	2.007235

```
# Find and display rules containing "other vegetables" on the right-hand side
all_rules[all_rules['RHS'].str.contains("other vegetables")]
```

	LHS	RHS	support	confidence	lift
6	root vegetables	other vegetables	0.047382	0.434701	2.246605
7	whipped/sour cream	other vegetables	0.028876	0.402837	2.081924
13	whole milk, root vegetables	other vegetables	0.023183	0.474012	2.449770